SEUNGWOO CHOI

ELECTRICAL AND ELECTRONIC ENGINEERING PHD CANDIDATE
QUANTUM SOFTWARE ENGINEER

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EDUCATION & EXPERIENCE

Mar 2022 – Feb 2028 (Expected)

Yonsei Univ. — **PhD Student**
Seoul, South Korea — Prof. Won Woo Ro
Electrical and Electronic Engineering

Jan 2025 – Jul 2026

Rutgers Univ. — **Visiting Scholar**
Piscataway, NJ, USA — Prof. Yipeng Huang
Computer Science

Mar 2016 – Feb 2022

Yonsei Univ. — **B.S. Graduate**
Seoul, South Korea — GPA: 3.64 / 4.3
Electrical and Electronic Engineering

SKILLS

Quantum SDK
Qiskit
QDK
Cirq
PennyLane

Circuit Simulator
qiskit-aer
qsimcirq
qulacs
QuEST

Software Development
Git
Shell Script
Python
C / C++

Hardware & Parallel
Verilog
CUDA

Languages
Korean
English

SUMMARY

- **Quantum Software Expert:** Specialized in quantum algorithms for chemistry, circuit simulation, circuit compilation, and QEC.
- **Systems Background:** Strong foundation in classical computer architecture applied to quantum software stack optimization.
- **Top-tier Publications:** Proven research excellence through publications in top-tier journals and conferences, including *HPCA*, *TETC*, and *ISCA*.
- **Industry Collaboration:** Experienced in industry-partnered research with *Samsung Research* and *Hyundai*.

SELECTED RESEARCH PROJECTS

[Jun 2026 – Present] **Ensemble-Guided Configuration Carryover for SQD**
- Proposed ranking carryover by the ensemble over all subsampled batches rather than the lowest-energy batch alone.
- Implemented on top of `qiskit-addon-sqd` and compared against the carryover rules of baseline SQD and TrimSQD.
- Evaluated robustness across trial states using LUCJ samples from real hardware together with simulated sample sets.

[Jun 2025 – Present] **Compilation-Aware Operator Selection for ADAPT-VQE**
- Incorporated compilation-induced cost — layout-induced two-qubit gates (NISQ) and magic-state-distillation-bound non-Clifford cost (EFTQC) — into the operator selection logic of ADAPT-VQE.
- Studied state-of-the-art ADAPT-VQE variants and implemented them on top of Qiskit's `AdaptVQE`.
- Studied the SABRE algorithm and gained hands-on experience in qubit layout and routing.

[May 2024 – Present] **RL-Driven Gate Fusion Optimization**
- Developed an RL agent to find the optimal gate fusion combinations to accelerate classical simulation.
- Analyzed and implemented gate fusion techniques from various simulators such as `qiskit-aer`, `qsimcirq`, and `qulacs`.
- Utilized DAG-based circuit representations to extract features for actual RL agent training.

[Oct 2023 – Dec 2024] **Accelerating UCCSD Circuit Simulation using Merged Gates**
- Accelerated circuit simulation by replacing patterns of `PauliEvolutionGate` blocks within the UCCSD ansatz with equivalent pseudo 1-qubit gates.
- Gained a holistic understanding of the entire molecular simulation pipeline, from chemical-to-qubit mapping to circuit implementation.
- Customized `qiskit-aer` code and executed VQE using the `qiskit` estimator.

[Mar 2023 – Apr 2024] **Memory-Efficient State-Vector Simulation using Polar Form Representation**
- Enhanced memory efficiency by storing complex coefficients as radius and phase instead of real and imaginary parts.
- Studied the principles and bottleneck of state-vector simulation through hands-on use of the `QuEST` simulator.
- Studied algorithms such as Shor, Grover, QAOA, and VQE to analyze and understand patterns in various benchmark circuits.

PUBLICATIONS

[HPCA '27] "**Compilation-Aware Adaptive Ansatz** for Near-Term and **Early Fault-Tolerant** Quantum Chemistry"
Incorporated compilation-induced cost into ADAPT-VQE operator selection across the NISQ and EFTQC regimes. (1st Author)

[QMI '26] "Layerwise Retraining and Freezing for **Multi-Angle QAOA**"
Proposed a round-robin layerwise training scheme with selective freezing for efficient Multi-Angle QAOA optimization. (4th Author)

[HPCA '26] "d'ArQ: A **QOC Framework** with Causality-Aware Grouping and Basis Selection"
Proposed a causality-aware grouping algorithm and basis selection to optimize pulse-level quantum control. (3rd Author)

[PRR '25] "Distribution-Adaptive Dynamic **Shot Optimization** for **Variational Quantum Algorithms**"
Developed a history based shot count prediction strategy to reduce VQA measurement cost. (4th Author)

[TETC '25] "q-Point: A Numeric Format for **Quantum Circuit Simulation** using Polar Form Complex Numbers"
Introduced a polar form numeric format to enhance memory efficiency in classical state-vector simulation. (1st Author)

[ISCA '25] "QR-Map: A Map-Based Approach to Quantum Circuit Abstraction for **Qubit Reuse Optimization**"
Designed a qubit reuse framework that abstracts circuits into maps to minimize required physical qubits. (3rd Author)

[CGO '25] "Qubit Movement-**Optimized Program Generation** on Zoned **Neutral Atom Processors**"
Developed circuit optimization techniques to minimize qubit movements on zoned neutral atom processors. (4th Author)

[ASP-DAC '25] "PIMutation: Exploring the Potential of Real PIM Architecture for **Quantum Circuit Simulation**"
Evaluated the effectiveness of Processing-In-Memory (PIM) architectures in accelerating quantum simulation tasks. (3rd Author)

[ICCD '24] "MOSQ: Accelerating **Classical Simulation** of **UCCSD Ansatz** Circuits using Merged Operation"
Accelerated UCCSD ansatz simulation by merging operations based on Pauli evolution patterns. (1st Author)

[ICCAD '24] "Barber: **Balancing Thermal Relaxation** Deviations of NISQ Programs by Exploiting Bit-Inverted Circuits"
Mitigated thermal relaxation errors by combining results from the original circuit and its bit flipped counterpart. (2nd Author)

[PACT '24] "**Recompiling QAOA Circuits** on Various Rotational Directions"
Developed a QAOA recompilation technique that changes rotation bases to reduce post compilation gate count. (3rd Author)

[DAC '23] "**Improving Fidelity** of Quantum Program by Independent Execution of Controlled Gates"
Improved fidelity by running controlled gate dependent subcircuits separately and combining measurement outcomes. (2nd Author)

PATENTS

[US Patent, Pending] "**METHOD FOR OPERATING A QUANTUM COMPUTING SYSTEM AND A QUANTUM COMPUTING SYSTEM**" (App. No. 18/966,013)

[KR Patent, Pending] "**SIMULATION METHOD FOR QUANTUM CIRCUIT AND SIMULATION DEVICE THEREOF**" (App. No. 10-2025-0161741)

[KR Patent, Pending] "**METHOD FOR OPERATING A QUANTUM COMPUTING SYSTEM AND A QUANTUM COMPUTING SYSTEM**" (App. No. 10-2024-0125804)

ACADEMIC SERVICE

Peer Review

[QMI '26] **Journal Reviewer**

[MICRO '26] **Artifact Evaluation Committee**

[ASPLOS '25] **Artifact Evaluation Committee**

Teaching Experience

[Fall 2023, Yonsei] **Embedded System Lab** (Graduate)

[Fall 2022, Yonsei] **Operating System** (Undergraduate)

[Spring 2022, Yonsei] **Electrical and Electronic Engineering Capstone Design** (Undergraduate)

OUTREACH & PRESENTATIONS

Quantum Fellowships & Mentoring

[Spring 2025] **IonQ Research Mentoring Program**

[Winter 2024] **IBM-Yonsei Quantum Challenge**

Selected Presentations

[Sep 2025, Rutgers] **SQD: Is VQE Being Replaced?**

[Aug 2025, Rutgers] **Introduction to LDPC-BB Code**

[Jan 2025, Yonsei] **Reinforcement Learning for Gate Merging Technique**

[Aug 2024, Yonsei] **VQE: From Computational Chemistry to Quantum Computing**

[Mar 2024, Yonsei] **Mitigating Memory Bottlenecks in Classical Simulation using Polar Format**

[Sep 2023, Yonsei] **Tutorial for QAOA: A Top-Down Approach**

[Mar 2023, Yonsei] **Meaning and Role of Phase in Quantum Computing**

[Sep 2022, Yonsei] **Quantum Circuit Simulation: Basic Concepts & What We Can Do About It**

[Sep 2021, Yonsei] **Understanding Shor's Algorithm**